

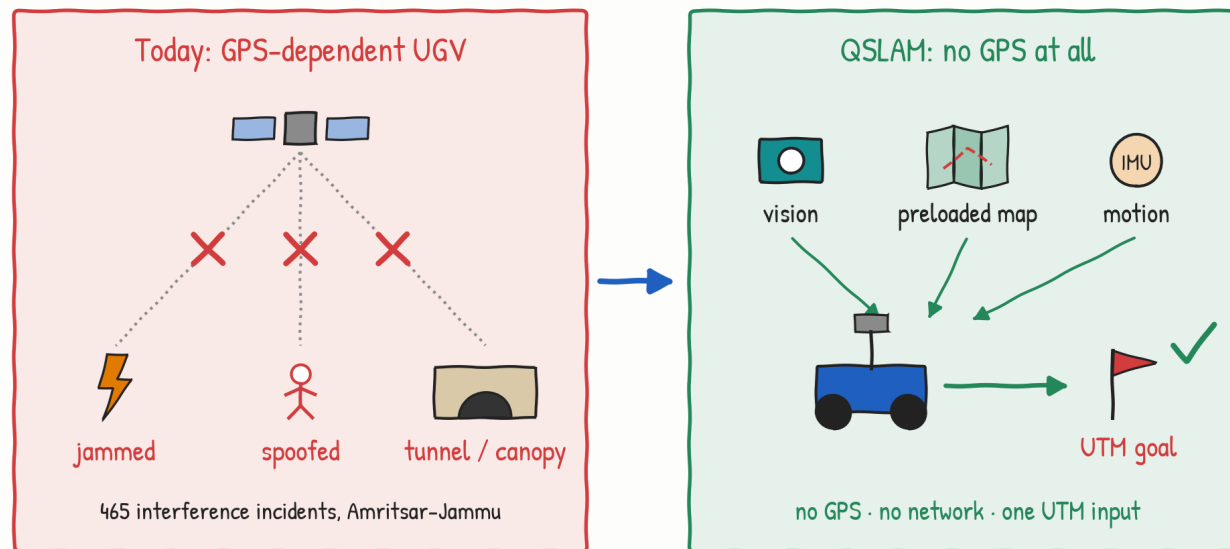
QSLAM

GPS-free, vision-based UGV navigation to a UTM goal

SIH 2026 · SIH26126 · Vision Based Autonomous Navigation for Unmanned Ground Vehicle for Outdoor environment
Smart Automation · Software · Team QSLAM (ID 14) · Bharat Electronics Limited

Where GPS fails – and QSLAM does not

1 · the problem



"No GPS. No network. Just vision, a map and a UTM goal."

1 **Solution**
what it is · how it solves the problem · what is new

2 **Technical approach**
frameworks · architecture · flowcharts

3 **Feasibility & viability**
measured results · risks & fixes

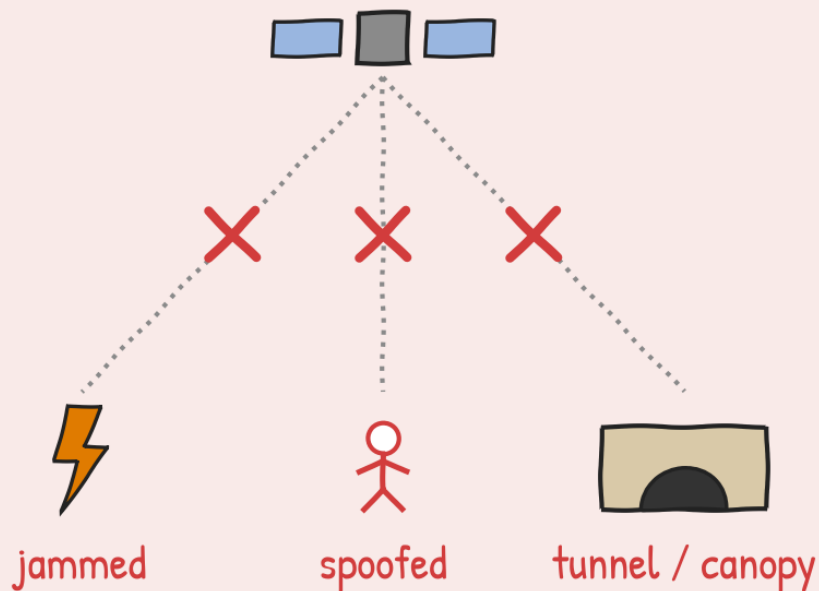
4 **Impact & benefits**
target users · benefits · roadmap

5 **Research & references**
papers · tools · links

Where GPS fails – and QSLAM does not

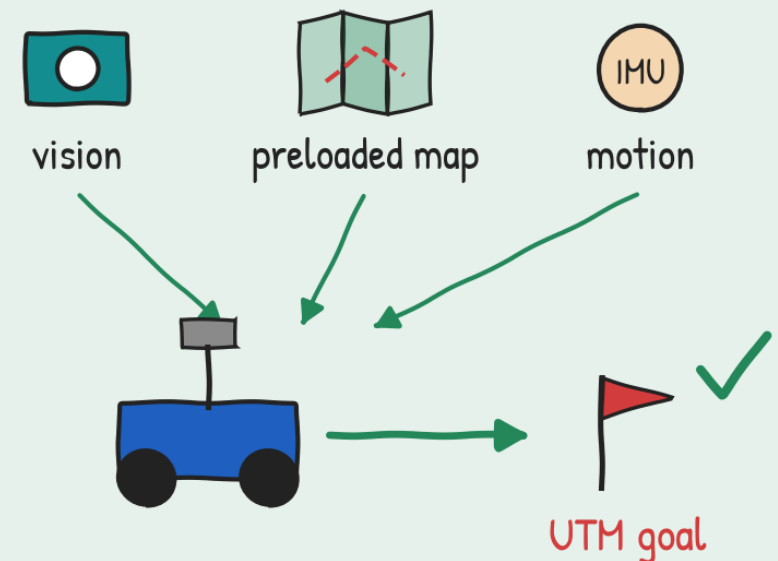
1 · the problem

Today: GPS-dependent UGV



465 interference incidents, Amritsar-Jammu

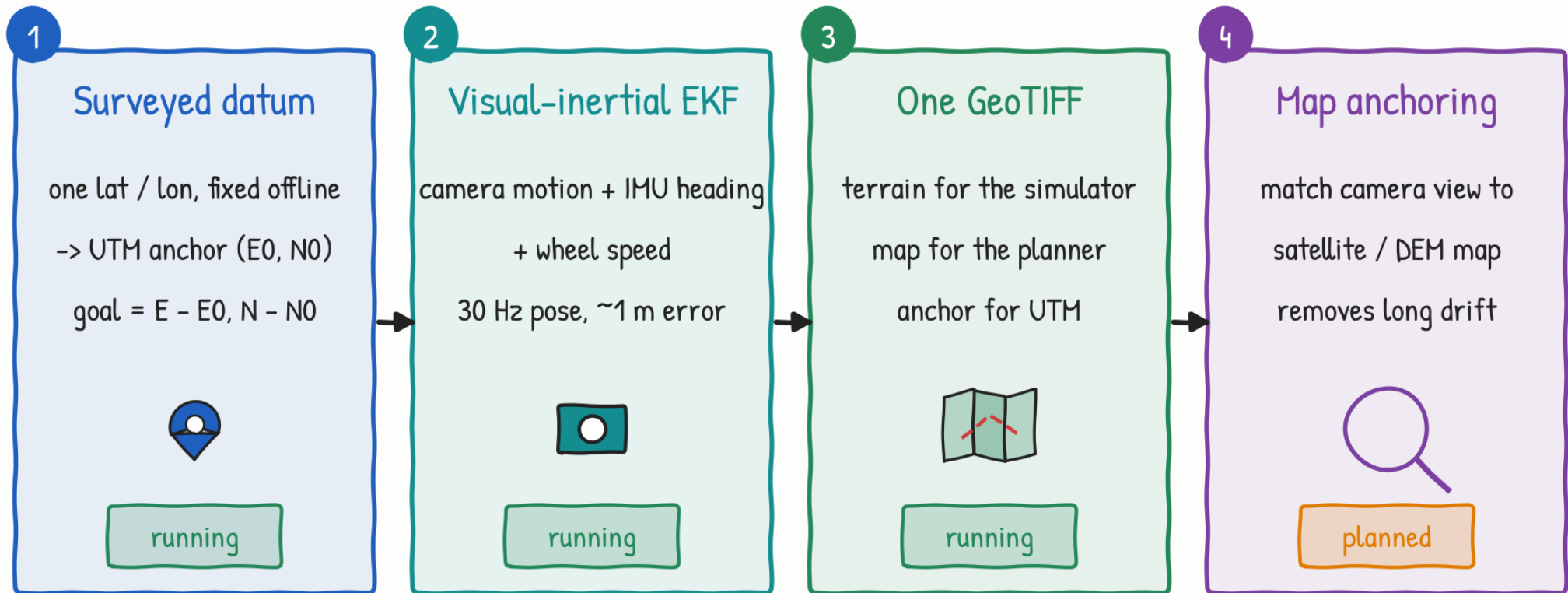
QSLAM: no GPS at all



no GPS · no network · one UTM input

How QSLAM knows where it is - without GPS

1 · the idea



1-3 run today · 4 plays the role GPS used to play

surveyed datum

visual-inertial EKF

one GeoTIFF

map anchoring

What is new

1 · innovation & uniqueness

0

Zero-GNSS by construction

GPS topics

no GPS topic, node or parameter

nothing to jam or spoof

0.000 mm

UTM goal without GPS

round-trip error

datum → anchor offline

run time = one subtraction

1

One GeoTIFF = one truth

source file

simulator terrain, planner map

and UTM anchor from one file

3

Prior + live + semantic

layers fused

DEM sees ditches cameras miss

lidar sees people, AI sees water

immune to jamming

exact UTM goal

map = world

prior + live + AI

Existing ways vs QSLAM

1 · how it addresses the problem

existing

limitation

QSLAM

GNSS / RTK-GPS

✗ jammed, spoofed, lost in tunnels → ✓

no receiver – immune

wheel odometry + gyro

✗ slip & drift: 10–70 m → ✓

EKF: VIO + heading, ~1 m

2-D lidar SLAM

✗ breaks on slopes after ~50 m → ✓

visual route + DEM prior

pure visual odometry

✗ drift grows, low texture → ✓

map anchoring + localizability

camera-only map

✗ misses ditches, no geo-frame → ✓

DEM shows ditches, goal in UTM

tele-operation

✗ radio link + operator → ✓

one UTM input, fully offline

no receiver

~1 m vs 10–70 m

sees ditches

fully autonomous

Technology stack

2 · frameworks

Geospatial

GeoTIFF · rasterio · pyproj · UTM 44N

Perception

SegFormer (RELLIS-3D / RUGD) · TensorRT

Navigation

Nav2: Smac A* · MPPI · behaviour tree

Localization

robot_localization EKF · Isaac ROS cuVSLAM

Middleware

ROS 2 Jazzy (LTS)

Simulation

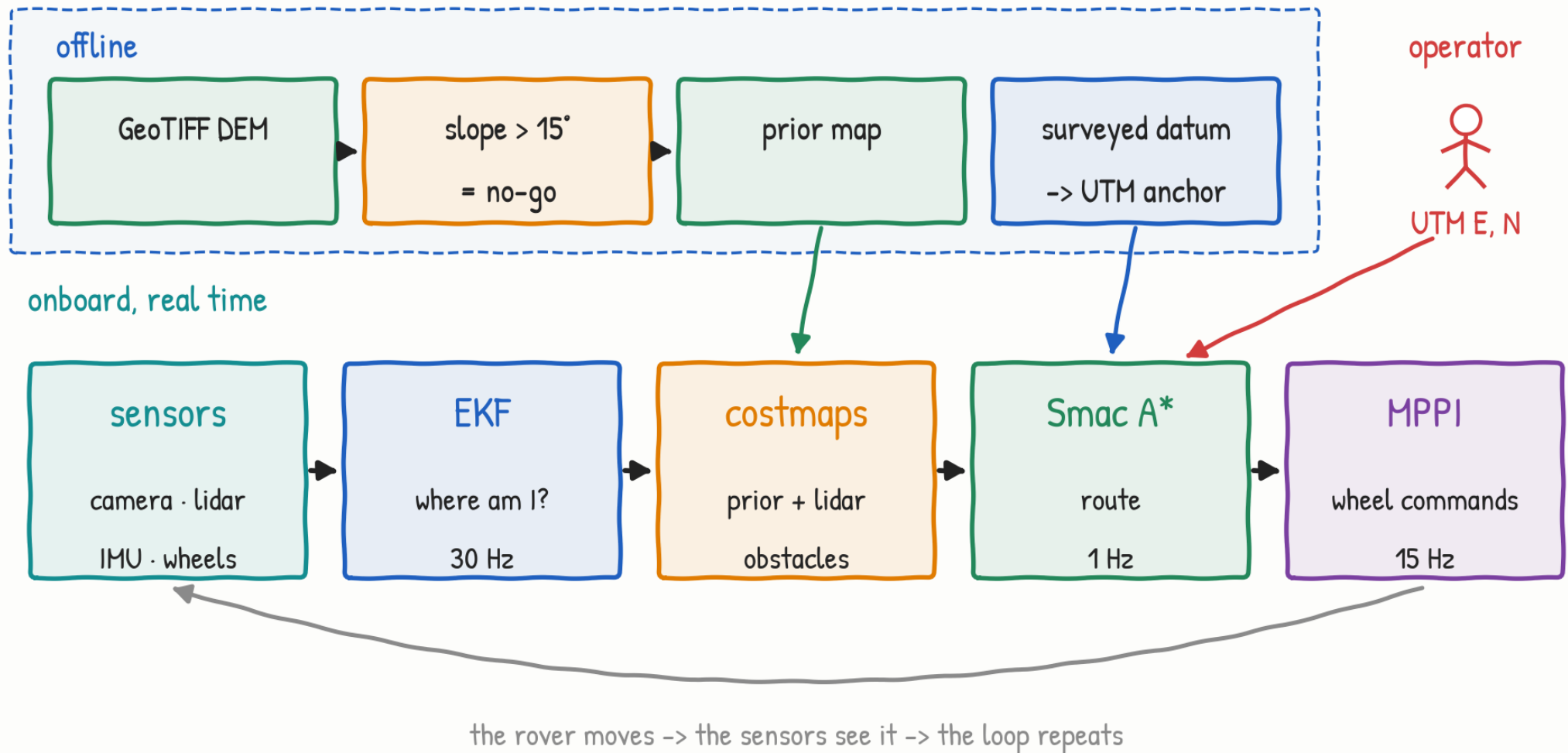
Isaac Sim 6 · Gazebo Harmonic · MuJoCo

Compute

RTX 4090 testbed -> Jetson AGX Orin onboard

System architecture

2 · offline preparation -> onboard loop



offline map prep

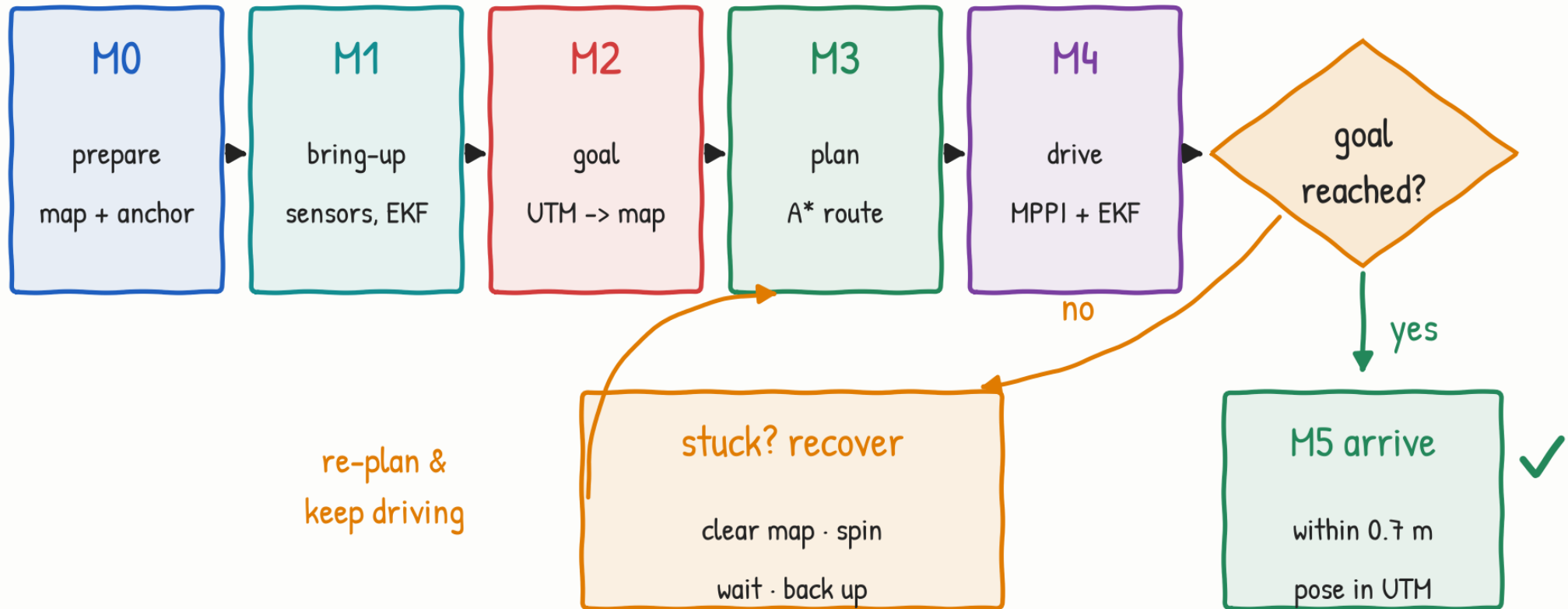
EKF 30 Hz

A* 1 Hz

MPPI 15 Hz

Mission flow

2 · methodology & process



M0 prepare

M1 bring-up

M2 goal

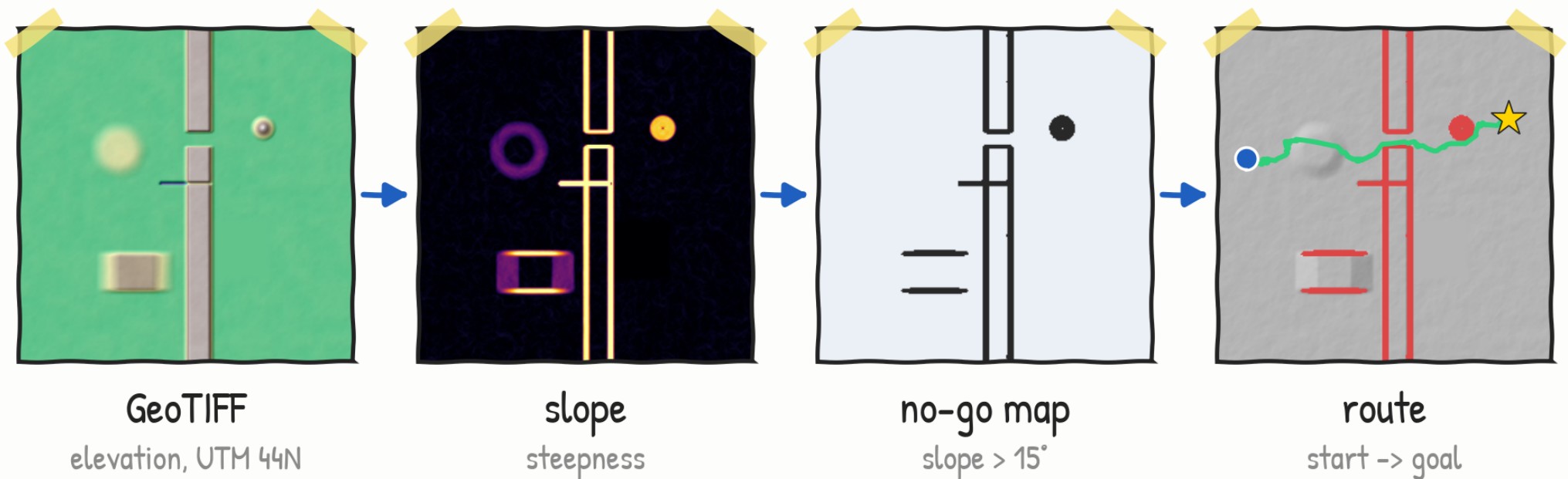
M3 plan

M4 drive

M5 arrive

From GeoTIFF to a drivable route

2 · map preparation



one file builds the world, the map and the UTM anchor

GeoTIFF

slope

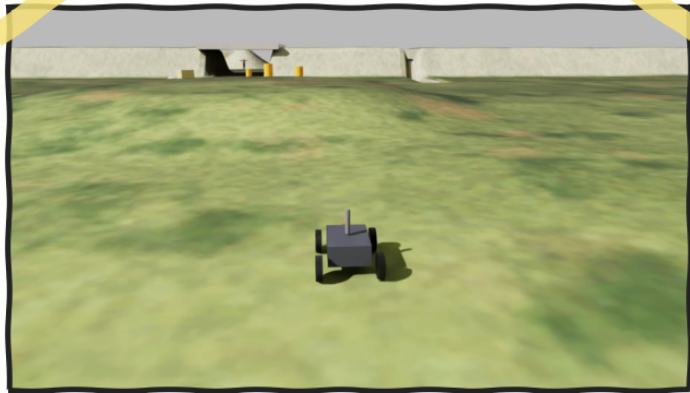
no-go map

route

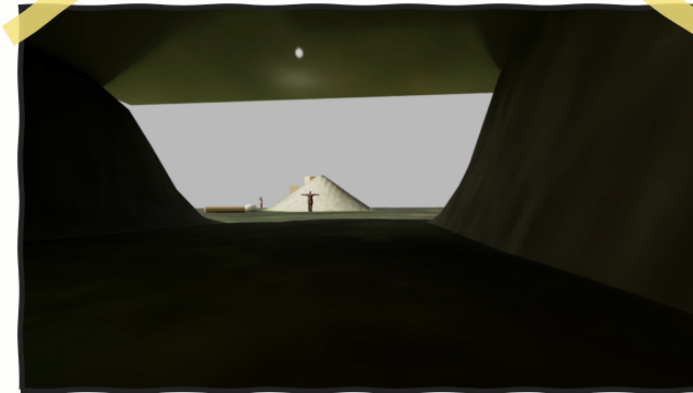
Recorded GPS-free mission (Isaac Sim)

2 · prototype

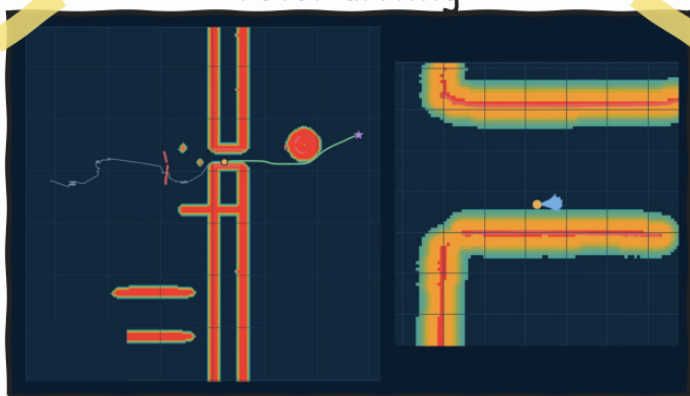
187 m driven · through the tunnel · goal reached 0.17 m away · no GPS



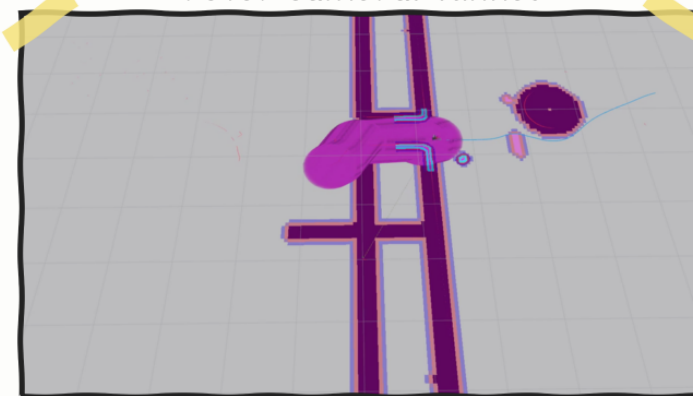
rover driving



rover camera: tunnel



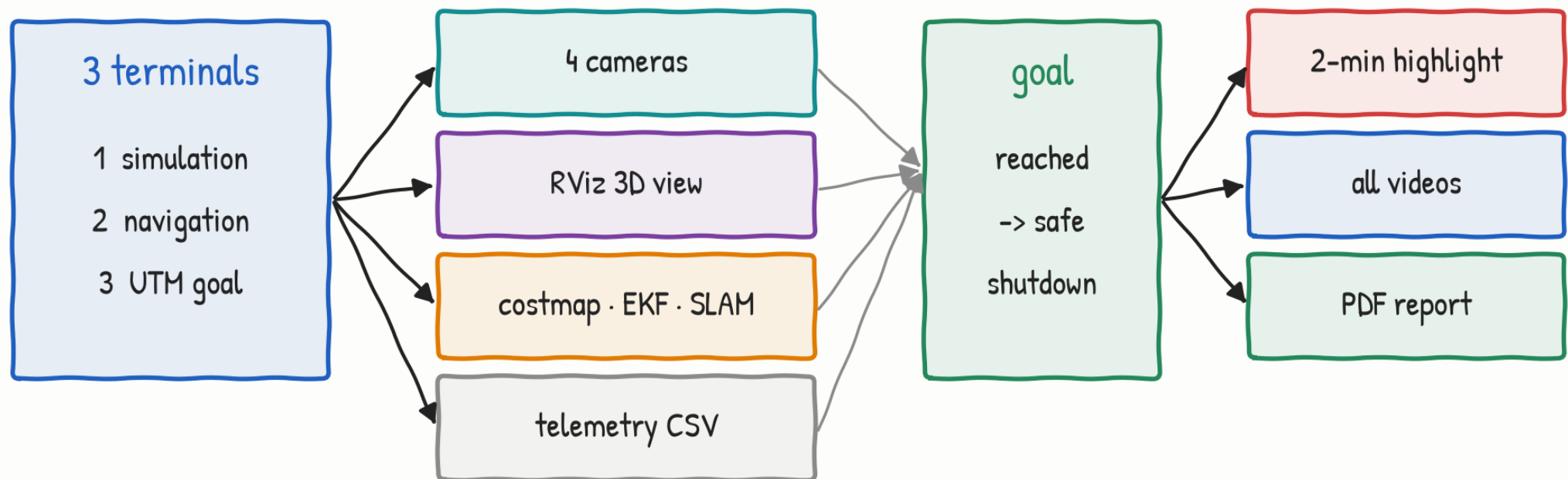
costmaps + route



RViz 3D view

Every run records itself

2 · evidence pipeline



pictures only – no words burned into any video
a stray Ctrl-C cannot corrupt the recordings

Measured, not assumed

3 · feasibility (Isaac Sim testbed)

4 / 4

missions reached the goal
through the tunnel

~1 m

EKF error vs 10-70 m
wheel odometry

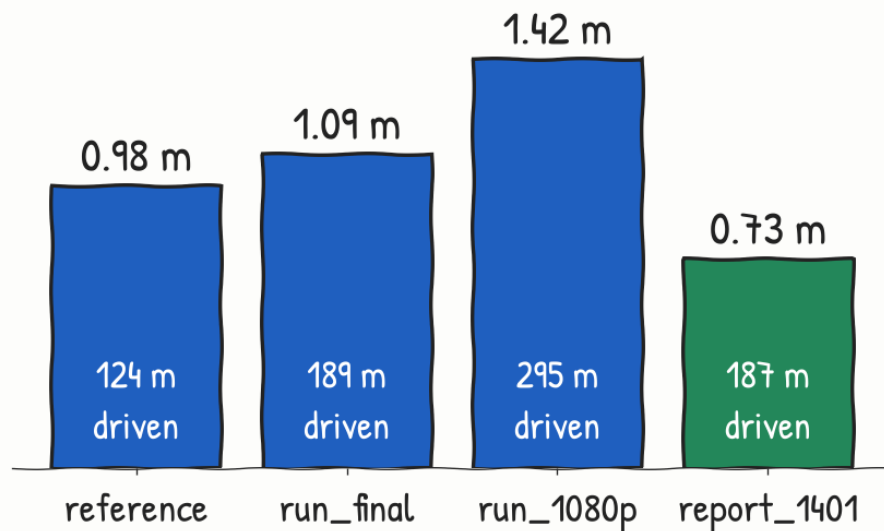
0.000 mm

UTM <-> map
round trip

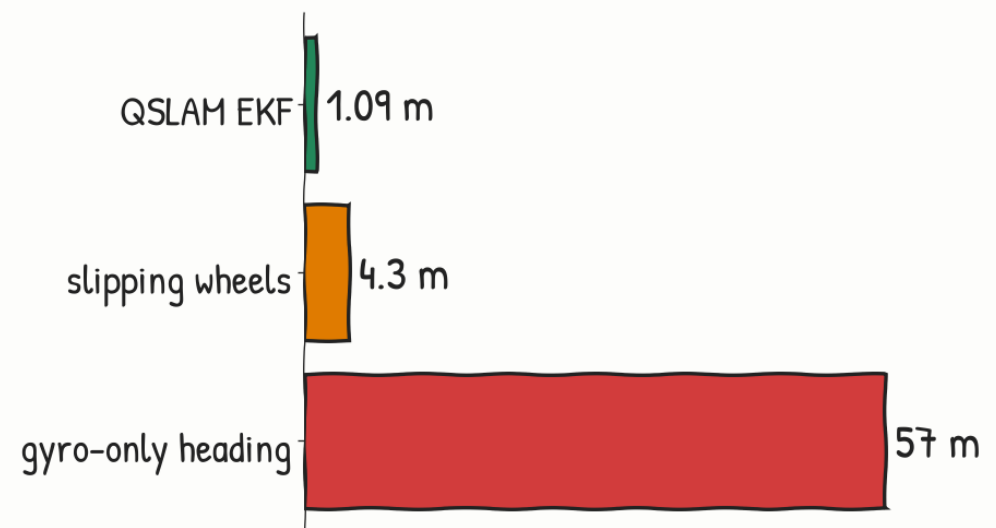
0 / 21

contacts with unmapped
obstacles (prototype)

final position error per run

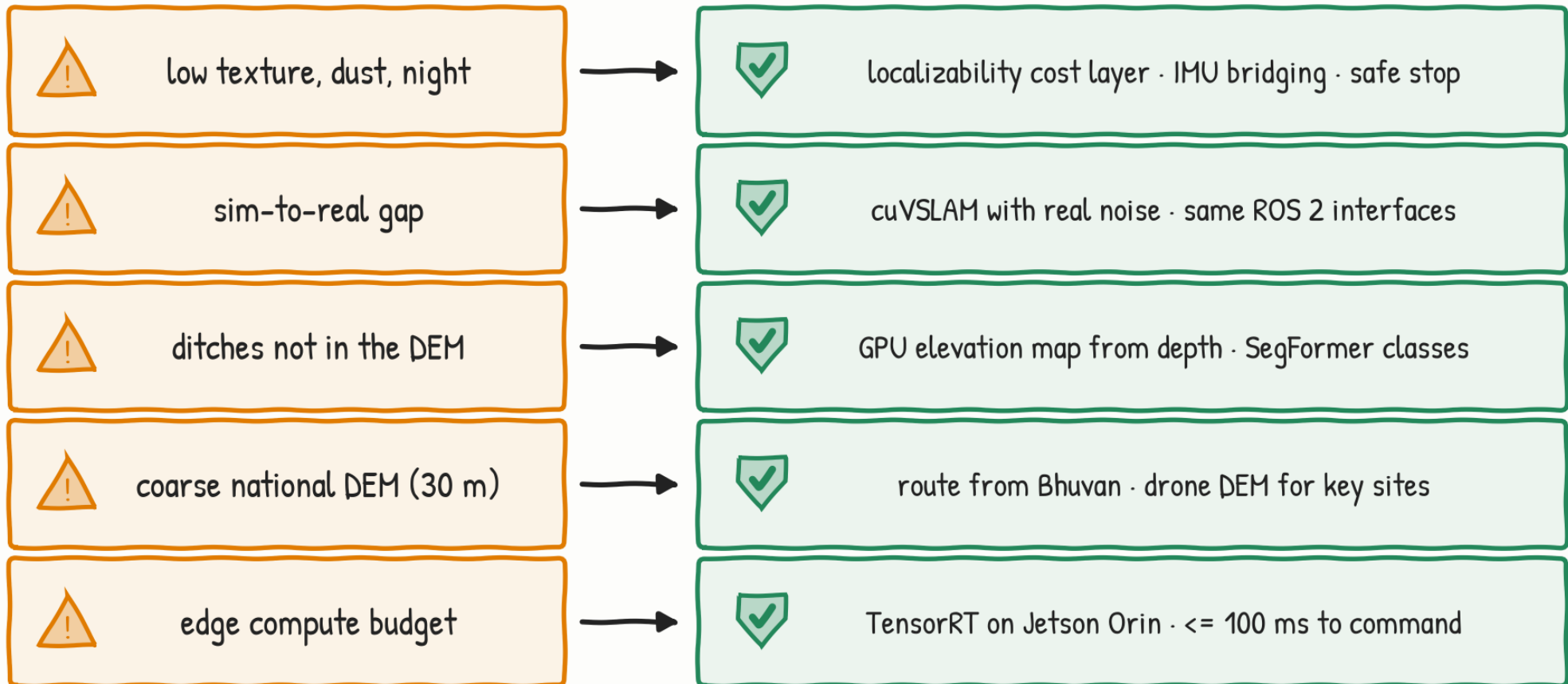


error after the same 189 m drive



Risks -> how we handle them

3 · challenges



Performance · safety · security · cost

3 · viability

real time

- ✓ MPPI 15 Hz
- ✓ EKF 30 Hz
- ✓ re-plan 1 Hz
- ✓ 0.5 s stop watchdog

safety

- ✓ 0.95 m obstacle margin
- ✓ off-site goals rejected
- ✓ recovery behaviours
- ✓ collision monitor (next)

security

- ✓ no GNSS to spoof
- ✓ no network in mission
- ✓ sensors only receive
- ✓ no remote attack path

cost

- ✓ Rs 0 software licences
- ✓ open source stack
- ✓ one COTS sensor kit
- ✓ Jetson AGX Orin

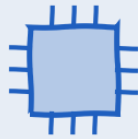
Who benefits

4 · impact on target audience



Army / CAPF

patrol & recce
without risking people
works under jamming



BEL

indigenous nav module
standard ROS 2 / Nav2
hardware-agnostic



NDRF / rescue

enters tunnels,
collapse & canopy zones
where GPS is lost



mining / tunnels

autonomous inspection
underground
no network needed

Army / CAPF

BEL

NDRF

mining

Benefits

4 · social · economic · environmental · strategic

social

keeps soldiers & rescuers
out of lead positions

economic

no licence cost
COTS sensors, not RTK / INS
Atmanirbhar

environmental

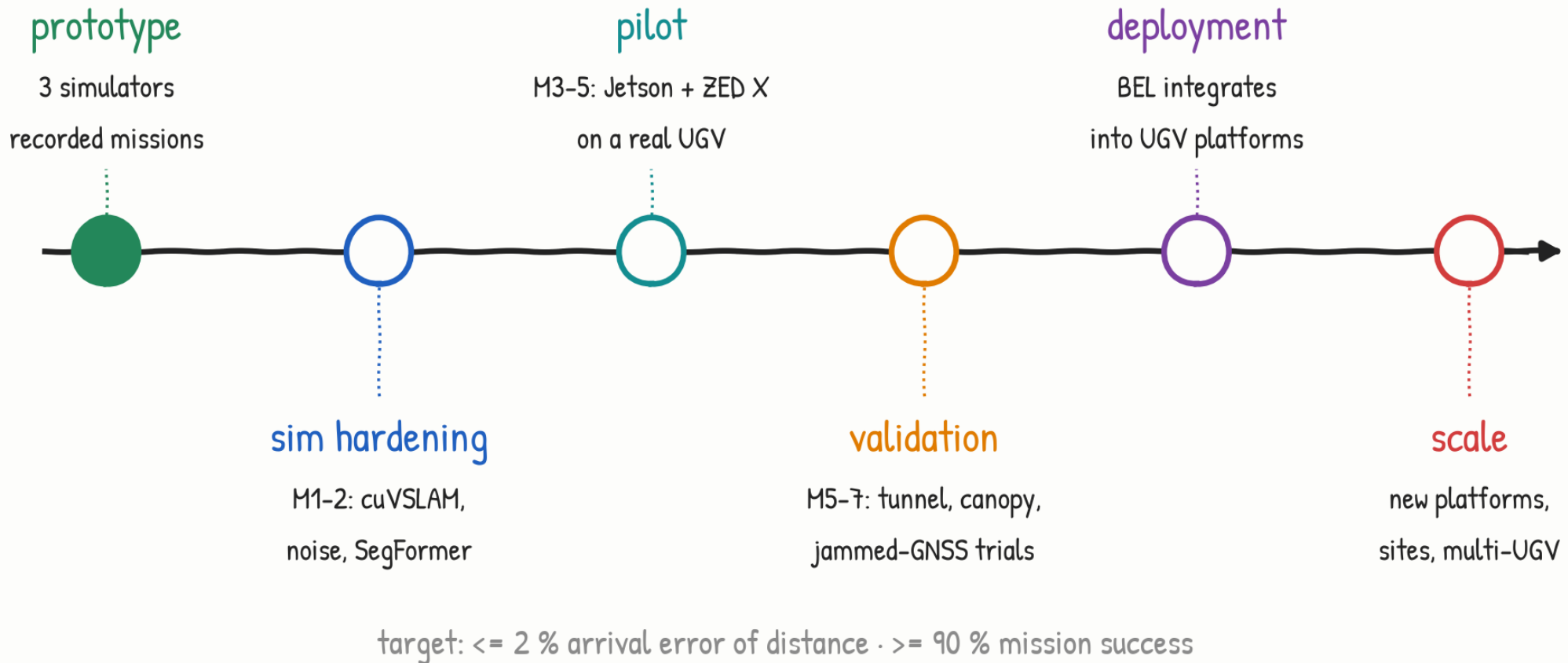
suits electric UGVs
fewer manned sorties

strategic

cannot be jammed,
spoofed or cut off

Roadmap

4 · prototype → scale



prototype done

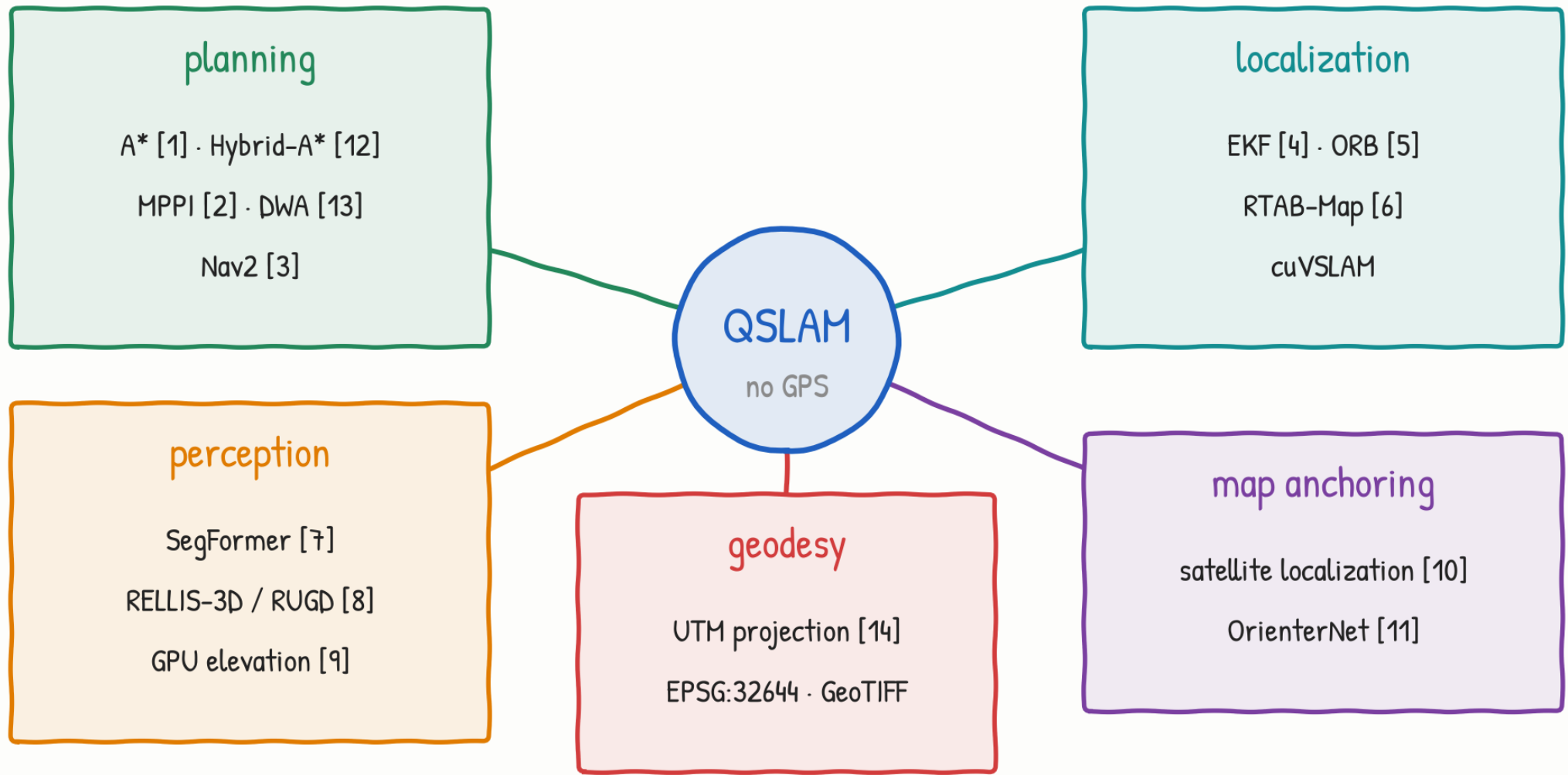
pilot

validation

deployment

Research behind each part

5 · research & references



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- tool** ROS 2 Jazzy
<https://docs.ros.org/en/jazzy/>
- tool** robot_localization
https://github.com/cra-ros-pkg/robot_localization
- tool** slam_toolbox
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- tool** Isaac ROS Visual SLAM (cuVSLAM)
https://nvidia-isaac-ros.github.io/repositories_and_packages/isaac_ros_visual_slam/index.html
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<https://bhuvan.nrsc.gov.in>
- tool** rasterio (GeoTIFF)
<https://rasterio.readthedocs.io>
- tool** pyproj
<https://pyproj4.github.io/pyproj/stable/>
- tool** EPSG:32644 - WGS 84 / UTM 44N
<https://epsg.io/32644>
- tool** Our code: SIH-26126
<https://github.com/kamalraj2772/SIH-26126>
- fact** Lok Sabha written reply (Mar 2025): 465 GPS interference / spoofing incidents, Amritsar-Jammu, Nov 2023 - Feb 2025
- fact** DGCA SOP (10 Nov 2025): real-time reporting of GPS spoofing / GNSS interference
- fact** SIH 2026 problem statement SIH26126 - Bharat Electronics Limited